A conditional proper-time volume response for prepared Einstein–scalar shell data

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Abstract

We calculate a conditional second variation of anchored proper-time volume for a compactly prepared perturbation of a five-dimensional Einstein–scalar spacetime with a pure-tension timelike shell. An explicit compact monopole particular solution fixes second initial data that were not numerically specified by an earlier weighted-only completion. A pure second-order gauge transformation removes a large spatial Lie-derivative term while retaining its scalar contribution. The complete five-component mean system is bounded by a fresh causal contraction estimate. Clock and volume-measure corrections give the proper scalar mean, and an exact intrinsic Gauss identity with a positive Green kernel gives the volume and expansion response. At the fixed proper time $\tau_* = (5/4)\rho_b 10^{-6}$, outward enclosures are $V_2/V_0 \in [-5.4520, -2.6521] \times 10^{16}$ and $\Theta_2 \in [-9.0208, -4.4458] \times 10^{21}$ in the stated model and parameter normalization. The conclusions assume the retained background, localized constraint inverse, smooth shell development, and source-identification results. They imply smaller volume and smaller positive expansion than the background for sufficiently small nonzero parameters in each fixed family; they give no effective parameter radius or physical radiation mechanism.

1 Problem, model, and conditional statement

Second-order observables depend on the nonlinear completion of first-order Cauchy data. A fixed first packet and zero initial tensor-transverse-traceless data do not determine the scalar second response. In particular, a weighted constraint construction whose radial weights have not been instantiated does not identify a unique numerical mean history. We address this problem by prescribing a compact mean particular solution and applying the retained constraint construction to the remaining source. This is a new preparation, not a reconstruction of unspecified old weights.

The bulk and one-copy outward shell equations are

$$\mathrm{Ric}_{AB} - \phi_A \phi_B - \tfrac{2}{3}\mathcal{V}(\phi) g_{AB} = 0, \qquad \Box_g \phi - \mathcal{V}'(\phi) = 0, \tag{1}$$

$$K^{\mathrm{out}}_{ab} = \tfrac{1}{6}\sigma(\phi)\gamma_{ab}, \qquad n_{\mathrm{out}}\phi = -\tfrac{1}{2}\sigma'(\phi). \tag{2}$$

Two identical interior copies use their respective outward normals. The retained model is

$$W(\phi) = 1 - \phi + \phi^3/3, \qquad \mathcal{V} = \tfrac{1}{2}(W')^2 - \tfrac{2}{3}W^2, \qquad \sigma = 2W + \delta(1 + c_\sigma \phi),$$
$$\delta = 10^{-3}, \qquad c_\sigma = 96714321425692/161839258930731. \tag{3}$$

The selected near-$\phi = 1$ regular background, with its retained shell data, is

$$g_0 = \mathrm{d}y^2 + \rho(y)^2[-\mathrm{d}t^2 + \cosh^2 t\,\kappa], \qquad a = \rho_y/\rho, \quad p = \phi_{0,y}, \tag{4}$$

where κ is the unit round metric on S^3. Put $\ell = 10^{-6}$, $x = t/\ell$, $r = z/\ell$, and $\mathrm{d}y = -\rho\,\mathrm{d}z$. Thus $\partial_r = -\ell\rho\partial_y$. The shell radius is ρ_b. The coordinate t is not proper time; on the background shell $\tau = \rho_b t$.

All subscript-2 quantities are second parameter derivatives:

$$g_\varepsilon = g_0 + \varepsilon q + \tfrac{1}{2}\varepsilon^2 k + o(\varepsilon^2), \qquad \phi_\varepsilon = \phi_0 + \varepsilon\eta + \tfrac{1}{2}\varepsilon^2 w + o(\varepsilon^2). \tag{5}$$

An angular mean uses probability measure on S^3. The first angular factor is $Y = n_1^2 - n_2^2$, so $\langle Y\rangle = 0$, $\langle Y^2\rangle = 1/6$, and $\Delta_\kappa Y = -8Y$.

The following are explicit analytic premises, not theorems reproved by the interval calculation: the regular background and its stated collar bounds; the invariant dotted-space weighted constraint inverse and its nonlinear implicit-function construction; a common fixed-shell development with the retained subsidiary and boundary uniqueness statements; its C^2 parameter dependence with the lower spacetime regularity needed for traces and observables; and identification of the actual first master and harmonic gauge fields with that family's first derivative. The companion technical records specify these premises and the fixed first seed [6]. The common solution slab contains the target with a strict time margin. For each sufficiently small fixed family, the anchored intrinsic normal foliation exists on the needed compact interval after qualitative parameter shrinkage. No numerical no-caustic or parameter-radius estimate is supplied.

Proposition 1 (Conditional response at a fixed proper clock)**.** *Under these premises, choose the explicit mean preparation of Section 2, keep the retained first packet, and define the anchored observables of Section 5. At $\tau_* = (5/4)\rho_b\ell$, their second derivatives obey*

$$\begin{aligned} M_2(\tau_*) &\in [1.1296, 1.9779]\times 10^{30},\\ v_2(\tau_*) := V_2(\tau_*)/V_0(\tau_*) &\in [-5.4520, -2.6521]\times 10^{16},\\ \Theta_2(\tau_*) &\in [-9.0208, -4.4458]\times 10^{21}. \end{aligned} \tag{6}$$

The intervals are outward decimal displays of exact rational enclosures. Every fixed admissible choice of the remaining radial weights has this same second mean response. This does not identify the full finite-parameter families or give a uniform admissible parameter interval over those choices.

2 An explicit compact second mean

Write ρ_0 for the background spatial volume density, distinct from ρ. With the action normalization $R/2 - |\mathrm{d}\phi|^2/2 - \mathcal{V}$, the initial first canonical data in a background orthonormal spatial frame are

$$\gamma_1 = \mathrm{diag}(-4u, 2u, 2u, 2u)Y, \quad \pi_1/\rho_0 = \mathrm{diag}(-3s/2, 0, 0, 0)Y, \quad \phi_1 = vY, \quad \varpi_1/\rho_0 = jY, \tag{7}$$

where $s = u_t/\rho$, $j = v_t/\rho$. The actual initial packet is

$$Q(z) = \exp\left(4 - \frac{\ell^2}{(z-\ell)(2\ell - z)}\right) \qquad (\ell < z < 2\ell), \tag{8}$$

$$\begin{aligned} &\alpha = (\rho/\rho_b)^{3/2}p/p_b, \quad \kappa_r = \alpha_z/\alpha = \rho(5a/2 - \mathcal{V}'/p),\\ &u = \alpha Q/\rho^2, \quad u_t = e^{-z}u, \quad v = \frac{3\alpha(Q_z + \kappa_r Q)}{\rho^3 p}, \quad v_t = \frac{3\alpha e^{-z}[Q_z + (\kappa_r - 1)Q]}{\rho^3 p}. \end{aligned}$$

Set $Q = 0$ elsewhere; its endpoints are flat. In particular $pv = -3(u_y + 2au)$ and $pj = -3(s_y + 3as)$.

Let Q_H and Q_M be the monopole Hessians of the canonical Hamiltonian constraint and the radial momentum constraint divided by ρ_0. Write the latter as Q_M^{mom}. Their explicit expressions are

$$\begin{aligned} Q_H = & -2u_y^2 - 24auu_y - 2uu_{yy} - (16a_y + 32a^2 + 8p^2/3)u^2 - 4u^2/\rho^2 - s^2 \\ & - (j^2 + v_y^2 + \mathcal{V}''v^2)/6 - (4/3)puv_y - (4/3)v^2/\rho^2, \\ Q_M^{\mathrm{mom}} = & -2su_y + j(v_y + 4pu)/3. \end{aligned} \tag{9}$$

These are Hessians, not coefficients already divided by two. If P is the linearized constraint operator, the required second equation is $Px_2 = -D^2\widehat{\mathcal{C}}[v_{\mathrm{seed}}, v_{\mathrm{seed}}]$.

With $y_L = y(2\ell) < y_U = y(\ell)$, define

$$I = \int_{y_L}^{y_U} \rho^4 Q_H \,\mathrm{d}y, \qquad J = \int_{y_L}^{y_U} \rho^2 Q \,\mathrm{d}y > 0, \qquad f = \frac{2I}{9J} Q, \tag{10}$$

$$C[f] = -\tfrac{3}{2}(f_{yy} + 3af_y) - \mathcal{V}f, \qquad \xi(y) = -\frac{\displaystyle\int_{y_L}^{y} \rho^4 (Q_H + C[f]) \,\mathrm{d}s}{\rho^4 p^2}. \tag{11}$$

The mean canonical datum B_{mean} is

$$\begin{aligned} h = f\gamma_0 + \mathcal{L}_{\xi\partial_y}\gamma_0, \qquad \phi_2 = \varpi_2 = 0, \\ \tau := \pi_2/\rho_0 = \mathrm{diag}(0, -Q_M^{\mathrm{mom}}/(6a), -Q_M^{\mathrm{mom}}/(6a), -Q_M^{\mathrm{mom}}/(6a)). \end{aligned} \tag{12}$$

The displayed metric components are $h_{yy} = f + 2\xi_y$ and $h_{\mathrm{angular}} = f + 2a\xi$.

The radial identities proving the construction are

$$\begin{aligned} & P_g(f\gamma_0) = C[f], && P_g(\mathcal{L}_{\xi\partial_y}\gamma_0) = \rho^{-4}(\rho^4 p^2 \xi)_y, \\ & \int \rho^4 C[f] \,\mathrm{d}y = -\tfrac{9}{2} \int \rho^2 f \,\mathrm{d}y, && P_m(C, D) = -2[C_y + 3a(C - D)]. \end{aligned} \tag{13}$$

Thus the primitive in (11) has zero total integral and vanishes on both exterior sides. Flatness gives a smooth compact extension, and (12) solves $PB_{\mathrm{mean}} = -P_0 D^2\widehat{\mathcal{C}}[v, v]$. The denominator bounds $\rho \geq 129$, $|p| \geq 3/25000$, and $a \geq 111/1000$ hold on the support. The sign and factor $-Q_M^{\mathrm{mom}}/(6a)$ are essential.

For completeness, let $\mathcal{A}$ be the retained radial weighted correction map and ω its constraint weight. Apply the same local implicit-function theorem to

$$x_\varepsilon = x_0 + \varepsilon v_{\mathrm{seed}} + \tfrac{1}{2}\varepsilon^2 B_{\mathrm{mean}} + \mathcal{A}U(\varepsilon), \qquad U(0) = U'(0) = 0. \tag{14}$$

The derivative in U is the original isomorphism $L = \omega^{-2}P\mathcal{A}$. Smooth compact polynomial trial terms only alter local constants. Haar averaging P_0 on the full tensor spaces commutes with the radial operators and weights, is bounded on the dotted closures, and maps the reflection-invariant domain into itself. Hence $LP_0U''(0) = 0$, so injectivity gives $P_0U''(0) = 0$. Individual rotations need not preserve that reflection-invariant domain. Consequently the exact curve has the prescribed second mean for every remaining admissible radial weight choice. Other angular second data and higher parameter derivatives are not thereby fixed.

3 A conditioned second harmonic jet

The canonical-to-physical map is

$$K_{ij} = \frac{2}{\rho_\gamma}\left(\gamma_{ia}\gamma_{jb}\pi^{ab} - \tfrac{1}{3}\gamma_{ij}\gamma_{ab}\pi^{ab}\right), \qquad \Pi = \varpi/\rho_\gamma, \qquad K = \tfrac{1}{2}\mathcal{L}_n\gamma. \tag{15}$$

At the time-symmetric background it gives $\langle K_{2,yy}\rangle = Q_M^{\rm mom}/(3a) + 6us$, $\langle K_{2,\rm angular}\rangle = -us$, and $\langle \Pi_2\rangle = -uj/3$. Choose initial lapse $N_\varepsilon = \rho(1+\varepsilon uY)$ and zero shift. For $R^A = g^{BC}\widehat{\Gamma}^A_{BC}$ with $\widehat{\Gamma} = \Gamma[g_0]$, the exact initial relative-harmonic completion is

$$\dot{N} = N^2\,\mathrm{tr}_\gamma K - N^3R^t, \qquad \dot{\beta}^I = N^2(\Gamma^I_\gamma - R^I) - N\gamma^{IJ}\partial_J N. \tag{16}$$

The perturbed inverse metric contracts the reference connection. Together with $\dot{\gamma} = 2NK$ and $\dot{\phi} = N\Pi$, this fixes all gauge velocities. The physical constraints then supply the vanishing initial normal subsidiary derivative through the reduced equations; those accelerations are not extra freely chosen Cauchy data.

The spatial term $2\xi_y$ can have size about 10^{39} in this preparation. Evolving the original angular trace requires cancellation against a large radial contribution in the shell boundary lift. We use a pure second generator Z, initially

$$(Z^y, Z^t, Z^y_t, Z^t_t) = (-\xi, 0, 0, a\xi), \qquad Z^{\rm angular} = 0, \tag{17}$$

and solve $(\Box_0 + \mathrm{Ric}_0)Z = 0$, with $Z^y = 0$ and $\partial_y Z^t = 0$ at the shell. These are the monopole normal Dirichlet and tangential Neumann rows. They are derived directly for the monopole vector; an angular-gradient amplitude is absent. The smooth compatible two-component problem has a standard reflected Volterra construction. The companion check bounds its contraction constant on the retained triangle by $3/250$.

The second transformation is

$$\bar{k} = k + \mathcal{L}_Z g_0, \qquad \bar{w} = w + Z\phi_0. \tag{18}$$

It removes the spatial Lie term and gives $\bar{\phi}_2 = -p\xi$. It leaves second canonical momenta unchanged because their background values vanish; naturality of (15) preserves the physical conversion. The chosen time derivative in (17) keeps the second lapse and shift values fixed. Omitting the new scalar displacement would change the physical data and violate the linearized constraint cancellation.

Use the background frame $e_y = \partial_y$, $e_0 = \rho^{-1}\partial_t$, $e_a = (\rho\cosh t)^{-1}E_a$, and mean order $\mathbf{U} = (K_0, L_0, N_0, J_0, w_0)$ for the $yy, y0, 00$, angular-isotropic, and scalar slots. The positive angular norm has Gram matrix $\mathrm{diag}(1,2,1,3,1)$. In this conditioned representative,

$$\mathbf{U}_0 = (f, 0, -u^2/3, f, -p\xi), \tag{19}$$

$$\mathbf{V}_0 := \partial_x\mathbf{U}|_0 = \ell\begin{pmatrix} 2\rho Q_M^{\rm mom}/(3a) + 32uu_t/3 \\ \rho[-f_y + af - 13uu_y/3 - au^2/3] \\ -2\rho Q_M^{\rm mom}/(3a) + 4uu_t/3 \\ -4uu_t/3 \\ 0 \end{pmatrix}, \tag{20}$$

$$\partial_r\mathbf{U}_0 = -\ell\rho(f_y, 0, -2uu_y/3, f_y, -p_y\xi - p\xi_y). \tag{21}$$

In particular N_0 is a covariant timelike frame component, with no additional sign flip. The g_{ty} velocity includes $2(\gamma_1)_{yy}\dot{\beta}^y_1$; the g_{tt} value includes the quadratic lapse square. The vanishing scalar time velocity is the exact cancellation $\rho\Pi_2 + 2N_1\Pi_1 = 0$.

This is a change of second jet of the same prepared physical family. The flow of Z at parameter $\varepsilon^2/2$ preserves the shell and its initial anchor. It need not be a nonlinear harmonic transformation for finite ε. Only $dC[\mathcal{L}_Z g_0] = 0$ is required for the second jet. The first field and all its quadratic sources remain unchanged. The retained parameter regularity permits the needed second-jet and trace operations; no new lossless Sobolev composition theorem is asserted.

4 The complete five-mean estimate

Rotational equivariance of the linearized background bulk and shell operators makes all five invariant slots a closed system, even for second data with additional angular components. Projecting the exact family's second equations and applying the retained mixed linear uniqueness identifies their mean with the solution from (19)–(21). The operator, physical shell rows, and normal gauge row are retained together:

$$\mathbf{U}_{xx} - \mathbf{U}_{rr} = A_0\mathbf{U}_x + B_0\mathbf{U}_r + D_0\mathbf{U} + F_0, \quad L_0|_{r=0} = 0, \quad P_N\mathbf{U}_r = M_0P_N\mathbf{U} + G_0. \tag{22}$$

Here P_N selects the other four slots. The actual forcing includes the reduced Einstein/scalar Hessians and all physical and gauge shell sources, not a selected-sign replacement.

Set $T = 5/4$, $\Delta = T - 1 = 1/4$, and

$$D_T = \{0 \le x \le T,\ 0 \le r \le T-x\}, \qquad \|\mathbf{U}\|_X = \max\left\{sup_{D_T}|\mathbf{U}|_H/3, \sup_{D_T}|\mathbf{U}_x|_H, \sup_{D_T}|\mathbf{U}_r|_H\right\}. \tag{23}$$

The mean initial data and first quadratic sources vanish in the flat wedge $x + r \le 1$. Thus the linear mean response vanishes there. This does not impose that same wedge on every finite-parameter nonlinear field. For compatible differences, integration from the wedge gives $|\mathbf{U}|_H \le \Delta\,\|\mathbf{U}\|_X$ and shell trace $|\mathbf{U}(x,0)|_H \le (x-1)_+\,\|\mathbf{U}\|_X$.

Write $\mathcal{K}$ for the reflected bulk Duhamel operator and $\mathcal{L}$ for the Robin lift. Their bounds are $\|\mathcal{K}F\|_X \le T\sup|F|_H$ and $\|\mathcal{L}G\|_X \le \sup|G|_H$. The retained coefficient caps $\|A_0\|, \|B_0\| \le 10^{-4}$, $\|D_0\| \le 10^{-6}$ and $\|M_0\| \le 1/2000$ yield

$$q = T(2\cdot 10^{-4} + \Delta 10^{-6}) + \Delta/2000 = \frac{1201}{3200000} = 0.0003753125 < 1. \tag{24}$$

If $\mathbf{B}$ is the reflected free wave for the *new* initial pair, the exact total mean satisfies $\mathbf{U} = \mathbf{B} + \mathcal{T}\mathbf{U} + \mathbf{S}$. The entire forcing history is charged once using

$$F_H = 17387513156573, \quad G_H = 3851400, \qquad S_X = TF_H + G_H = \frac{86937581188465}{4}. \tag{25}$$

For new initial norm caps $D_0^{\text{init}}, D_r^{\text{init}}, D_v^{\text{init}}$,

$$B_X \le \max\{(D_0^{\text{init}} + TD_v^{\text{init}})/3, D_r^{\text{init}} + D_v^{\text{init}}\}, \qquad \|\mathbf{U} - \mathbf{B}\|_X \le E_X := \frac{qB_X + S_X}{1-q}. \tag{26}$$

This bounds all feedback iterations and all internal mean couplings. No old total cap, initial accuracy allocation, or truncated-iteration tail is substituted. Whole-cell interval evaluation with flat-tail treatment gives the outward displays

Quantity	Certified upper display
D_0^{init}	1.9944×10^{30}
D_r^{init}	1.7478×10^{31}
D_v^{init}	2.7533×10^{22}
B_X	1.7478×10^{31}
E_X	6.5620×10^{27}

The large coefficients reflect the chosen parameter normalization; they are not finite-amplitude field values. The construction bounds a continuous response rather than numerically time-stepping a full field.

5 Proper scalar mean and intrinsic volume

Anchor the shell's intrinsic future normal-geodesic foliation at its registered initial spatial section. Let $h_\varepsilon(\tau)$ be the induced spatial metric in that foliation and define

$$V_\varepsilon(\tau) = \int_{S^3} \mathrm{d}\mu_{h_\varepsilon}, \qquad \Theta_\varepsilon = \partial_\tau \log V_\varepsilon, \qquad M_\varepsilon = V_\varepsilon^{-1} \int_{S^3} \phi_\varepsilon \, \mathrm{d}\mu_{h_\varepsilon}. \tag{27}$$

The observers are intrinsic shell geodesics, not an assumed material flow or ambient geodesic congruence. The background has $A(\tau) = \rho_b \cosh(\tau/\rho_b)$, $H = A_\tau/A$, and $V_0 = 2\pi^2 A^3$. All first integrated coefficients vanish by the quadrupole mean.

To exhibit the required first corrections, write the first shell frame metric as $m_{00} = -2XY$, $m_{0a} = VD_aY$, and $m_{ab} = 2(ZY\kappa_{ab} + dD_aD_bY)$, and write the first scalar as ψY. Here V is a first metric amplitude only in these formulas, not the shell volume. Set $C_\tau = X$, $C(0) = 0$, and $E = Z - HC$. The normalized proper mean is

$$M_2 = w_0 - C\psi_\tau/3 + (E - 8d/3)\psi. \tag{28}$$

The last term is the perturbed measure correction. The d contribution is retained. Angular transport cancels only after scalar pullback and measure are combined, not by dropping both indiscriminately.

Let Q_{corr} bound the last two terms of (28). The actual first shell two-jet is enclosed on the whole interval $1 \le x \le T$, giving $Q_{\mathrm{corr}} < 3.7987$. Even reflection and zero initial scalar velocity give $B_w(x,0) = w_{\mathrm{init}}(x) = -p\xi|_{r=x}$. Hence

$$|M_2(x) - w_{\mathrm{init}}(x)| \le (x-1)E_X + Q_{\mathrm{corr}} \quad (1 \le x \le T). \tag{29}$$

Its linearly weighted and unweighted history error bounds are, respectively, $\Delta^3 E_X/6 + \Delta^2 Q_{\mathrm{corr}}/2$ and $\Delta^2 E_X/2 + \Delta Q_{\mathrm{corr}}$. Exact positive cell weights integrate the continuous initial scalar boxes. No assumption of pointwise positivity at every arbitrarily early contact time enters these history bounds.

The exact shell Gauss contraction of (1)–(2) is

$$R(\gamma) = (D\phi)^2 + F(\phi), \qquad F = 2\mathcal{V} - \tfrac{1}{4}(\sigma')^2 + \tfrac{1}{3}\sigma^2. \tag{30}$$

This standard geometric identity removes the electric Weyl trace; it does not determine every induced tensor equation. Define $v_2 = V_2/V_0$. The volume-weighted second identity in the anchored proper-time foliation is

$$(v_2)_{\tau\tau} + 4H(v_2)_\tau - 2v_2/A^2 = \tfrac{1}{2}F'M_2 + Q_1, \qquad v_2(0) = (v_2)_\tau(0) = 0, \tag{31}$$

$$\begin{aligned} Q_1 = {} & F''\psi^2/12 - \psi_\tau^2/6 + 4\psi^2/(3A^2) - 5E^2/(3A^2) \\ & + E_\tau^2 - 16E_\tau D_\tau/3 + 8D_\tau^2/3. \end{aligned} \tag{32}$$

Here D is the synchronous angular Hessian coefficient, with $D_\tau = d_\tau - V/A - C/A^2$. Every term is retained and bounded in absolute value; the resulting cap is $|Q_1| < 7.8317 \times 10^{19}$. In (31) the scalar source is the full proper mean, not the coordinate coefficient w_0.

One way to check the curvature coefficient is to vary the functional $\mathcal{J}(h) = \int R_3 \mathrm{d}\mu - (2/A^2)\int \mathrm{d}\mu$ at the round sphere. It is critical. For the coordinate metric variation $h_1 = 2A^2(EY\kappa + D\,\mathrm{Hess}\,Y)$, its normalized Hessian is $V_0^{-1}D^2\mathcal{J}[h_1,h_1] = 10E^2/(3A^2)$; a pure spatial diffeomorphism contributes zero. The weighted second spatial curvature is therefore $-4v_2/A^2 + 10E^2/(3A^2)$. The temporal part contributes $2(v_2)_{\tau\tau} + 8H(v_2)_\tau + 2\langle \mathrm{tr}\,K_1^2 - \theta_1^2\rangle$, with

$$\left\langle \mathrm{tr}\,K_1^2 - \theta_1^2 \right\rangle = -E_\tau^2 + 16E_\tau D_\tau/3 - 8D_\tau^2/3. \tag{33}$$

Combining this with $\langle (D\phi)^2\rangle_2 = -\psi_\tau^2/3 + 8\psi^2/(3A^2)$ proves (31)–(32), including the derivative convention.

In scaled time the exact Green kernel is

$$k(T,x) = \frac{\sinh(2\ell(T-x))}{2\ell}\frac{\cosh^2(\ell x)}{\cosh^2(\ell T)}. \tag{34}$$

It follows from the homogeneous solutions $\tanh(\ell x)$ and $1+\tanh^2(\ell x)$. It is positive, with $0 \le k(T,x) \le T-x$. On the interval used here, $|k/(T-x)-1| < 10^{-11}$ and $|k_T - 1| < 10^{-11}$, interpreted continuously at the endpoint. These uniform errors are charged to the history bounds. Consequently

$$v_2(T) = (\rho_b \ell)^2 \int_1^T k(T,x)[F' M_2(x)/2 + Q_1(x)]\,\mathrm{d}x, \tag{35}$$

$$\Theta_2(T) = (v_2)_\tau(T) = \rho_b \ell \int_1^T k_T(T,x)[F' M_2(x)/2 + Q_1(x)]\,\mathrm{d}x. \tag{36}$$

Here F' is evaluated at the constant background shell scalar. Its retained interval is negative. Equations (29) and (35)–(36), with the full source budgets, give (6). In particular Θ_2 is the first proper-time derivative of v_2, not an acceleration.

6 Interpretation, verification, and relation to prior work

At the chosen positive time the background expansion is

$$\Theta_0(\tau_*) = \frac{3}{\rho_b}\tanh(\ell T) > 0. \tag{37}$$

Because the first integrated coefficients vanish, C^2 parameter dependence gives, at this same anchored proper clock,

$$\frac{V_\varepsilon}{V_0} = 1 + \tfrac{1}{2}\varepsilon^2 v_2 + o(\varepsilon^2), \qquad \Theta_\varepsilon = \Theta_0 + \tfrac{1}{2}\varepsilon^2\Theta_2 + o(\varepsilon^2). \tag{38}$$

The strict signs in (6) therefore imply, for each fixed admissible family and sufficiently small nonzero ε of either sign,

$$0 < V_\varepsilon(\tau_*)/V_0(\tau_*) < 1, \qquad 0 < \Theta_\varepsilon(\tau_*) < \Theta_0(\tau_*). \tag{39}$$

No particular nonzero ε is certified. The allowed interval depends on the family and its remainder control, and need not be uniform in the remaining weights. The background is still expanding in (39); a negative second change is not total contraction.

The numerical statements use whole-cell outward interval arithmetic and exact rational endpoints, including the packet's flat tails. The initial campaign uses the registered $N = 1024$ base grid, with 994 recorded cells: 992 interior cells and two grouped flat-tail regions. The conditioned adapter is independently checked against every supplied cell, and the complete source history enters once through (25). Normal and optimized runs agree exactly. A separate reconstruction of the response budget matches all three final rational intervals without importing the response producer. These checks authenticate and audit the stated finite calculation; they do not constitute proofs of the retained analytic existence assumptions or external peer review. The accompanying materials provide the proofs, arithmetic adapters, source pins, raw boxes, and canonical reports [6].

The calculation uses established geometric and analytic machinery. Bruni et al. [1] give the second-order gauge law, including its factorial convention and first-generator cross terms; Mars [2] treats perturbations of hypersurfaces and the distinction between spacetime gauge and hypersurface identification. Transport of the shell and its anchor is necessary here. The Gauss–Codazzi projection used in brane geometry [3] is standard; constant-tension or constant-bulk matter reductions cannot be imported unchanged into (30). Buchert's averaging identities [4, 5] establish the expansion-variance/shear combination and the averaging commutator. Their dust or fluid evolution assumptions are not imposed on our intrinsically geodesic shell observers. The reflected-kernel contraction and Green-function calculation are standard linear methods.

The contribution is the model-specific connection between an explicit compact second mean, its conditioned five-field response, and these anchored observable enclosures. This is a focused comparison with those primary frameworks, not an exhaustive novelty survey or a new general implicit-function theorem, averaging law, or Einstein equation. The uncomputed higher angular sectors are not needed for this invariant projection, but no complete angular waveform is supplied. The large second coefficients have no SI calibration here. No quantitative finite-amplitude continuation, hot-matter or radiation history, physical packet-generation mechanism, or Big-Bang-origin conclusion follows.

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